Summary Report

Data Engineering & Big Data Specialist Blackbucks Long Term Internship 2026

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Internship Overview:

Total Duration: 4Months (16 Weeks)
Total Learning Hours: Up to 240 Hours
Delivery Mode: Hybrid (Live + Recorded + Platform-based Learning)
Learning Platform: Blackbucks TaPTaP LMS
Project : Individual Project selection
Communication Mode: Domain-wise WhatsApp Groups

Program Objectives

The **Blackbucks Skill Internship Program 2026** aims to:

1. Develop **industry-ready skills** through structured, domain-specific training.
2. Provide **hands-on project experience** and mentor-driven guidance.
3. Strengthen **placement and employability readiness** through mock tests and continuous evaluation.
4. Offer exposure to **modern technologies and workflows** in real-world problem-solving.
5. Enable participants to graduate with a **project portfolio and certification** validating 240 hours of learning.

Program Structure

Pre-Program Phase (Week 01 – 04)

Objective: Prepare interns for core learning and professional readiness.
Activities:

- Daily **Placement & Employability Practice Papers**
- **Pre-Assessment Test**
- **Prerequisite Courses & Recorded Content**
- **Orientation Session (2 Hours)** introducing mentors, LMS, and domains.

Core Internship Phase (Week 05 – 12)

Objective: Strengthen technical and practical expertise.
Weekly Pattern:

- 3 Live Sessions (2 hrs each)
- 1 Assignment
- 2 Assessments
- 1 Employability Test

Recap & Review Phase (Week 13 – 14)

Objective: Revise all domain concepts and prepare for project work.

- Mentor-led recap sessions
- Domain-wise revision & short tests

Project Implementation Phase (Week 15)

Objective: Apply learning through a guided, mentor-supported project.
Details:

- 2 Mentor Support Sessions per week (6 total)
- Project divided into 5–6 steps (each step: 3 days)
- Weekly placement and employability tests continue

Final Phase (Week 16)

Objective: Assess final performance and close the internship.

- **Grand Test – Part 01 & 02 (Total 5 Hours)**
- **Project Review & Certification Ceremony**

About the Blackbucks

The Blackbucks: Pioneering Campus Success

The Blackbucks is India's premier **Campus Success Platform**, committed to transforming higher education and placement ecosystems. As a trusted partner of top **Higher Education Institutions (HEIs)**, Blackbucks provides a robust suite of technology-driven solutions designed to empower students through rigorous training, structured practice, and successful placement outcomes.

Vision: Empowering Students for a Brighter Tomorrow

At The Blackbucks, we believe in **leveraging technology to unlock the full potential** of every student. Our mission is to bridge the gap between academic learning and real-world careers by offering comprehensive support systems that align education with employability.

Key Offerings

- **TaPTaP Platform** – Advanced learning and placement tracking system
- **VPL Platform** - College domain wise labs
- **BBX Swift** – Personalized learning engine for upskilling and certification
- **University Certification Programs** – Job-linked career pathways integrated with higher education

Together, these tools provide a **360° learning and placement experience**, tailored for diverse academic backgrounds.

12 Years of Impact & Innovation

Founded in **2013**, The Blackbucks began with a revolutionary idea: *"Technology for Students."* Over the past decade, we've grown into a world-class technology company with a track record of innovation, impact, and measurable success in student outcomes.

Recognized Leaders in Post-Graduation Careers

Our **Post Graduation Career Programs**, run in collaboration with the **Government of Andhra Pradesh**, have been **awarded for seven consecutive years**, showcasing our excellence in job-linked education and guaranteed placement pathways. We take pride in offering cutting-edge training in:

- Software & Product Development
- Artificial Intelligence & Machine Learning
- Data Science & Emerging Technologies
- Cybersecurity, MBA, and Entrepreneurship Tracks

Why Choose The Blackbucks?

- **10+ years of expertise** in student-centered ed-tech innovation
- **Guaranteed placement programs** across technical and management domains
- A growing network of **200+ partner colleges and 1000+ recruiters**
- Proven success with **tens of thousands of student placements**

Join Us: Where Learning Meets Opportunity

Whether you're a **student seeking career transformation**, an **educational institution aiming to boost placement performance**, or a **recruiter looking for top talent**, The Blackbucks is your destination for success.

Company Snapshot

- **Website:** www.theblackbucks.com
- **Industry:** Technology, Information & Internet
- **Headquarters:** Hyderabad, Telangana, India
- **Company Size:** 51–200 employees
- **Founded:** 2013
- **LinkedIn Presence:** 143+ professionals currently associated

Specialties

IT & Software | Product Development | Engineering | Electronics | AI/ML | Data Science | Emerging Technologies | Higher Education | Career Services | Entrepreneurship | Jobs & Placements

Course Syllabus :

Data Engineering & Big Data Specialist – 10 Module Syllabus

Module 1: Introduction to Data Engineering

- Overview of Data Engineering & Roles
- Big Data Lifecycle
- Batch vs Streaming Architectures
- Data Engineering Use Cases

Module 2: Data Storage & Formats

- Structured vs Unstructured Data
- Data Formats (CSV, JSON, Parquet, Avro)
- Data Schema Concepts
- Data Storage Systems Overview

Module 3: ETL & Data Pipeline Basics

- Extract–Transform–Load (ETL) Concepts
- Data Cleaning Techniques
- Data Pipeline Architecture
- Introduction to Workflow Automation

Module 4: Big Data Foundations

- 4 Vs of Big Data
- Distributed Systems Concepts
- Hadoop Ecosystem Overview
- Big Data Processing Models

Module 5: Hadoop & Data Warehousing

- HDFS Architecture & Replication
- MapReduce Workflow
- Hive Data Warehousing & HiveQL
- HBase for NoSQL Storage

Module 6: Apache Spark Fundamentals

- Spark Architecture & Components
- RDDs, Transformations & Actions
- Spark DataFrames & Spark SQL

- Query Optimization (Catalyst & Tungsten)

Module 7: Real-Time Data Processing

- Linear, Ridge & Lasso Regression
- Classification Models
- Model Selection & Validation
- Business Use Cases

Module 8: Data Orchestration & Data Lakes

- Apache Airflow & DAG Workflows
- Task Scheduling & Operators
- Data Lake Architecture
- Partitioning & Data Organization

Module 9: Cloud Data Engineering & Governance

- Cloud Data Warehouses (Redshift, BigQuery, Synapse)
- Monitoring & Logging (ELK Stack)
- Data Governance & Security
- Compliance & Data Privacy

Module 10: Advanced Pipelines & Capstone Project

- ETL Pipelines using PySpark
- ML Pipeline Integration & Feature Engineering
- Mini Project: ETL + Streaming Pipeline
- Capstone Presentation & Industry Readiness

Pre-Program Phase- 01 (Week 01 – 04)

Monthly Summary Report – December 2025

Internship Duration Covered

Month: December 2025 – Early January 2026
Program: Blackbucks Long Term Internship (LT 2026)
Focus Areas: Pre-requisite Skill Building, Company-Based Evaluation, Employability Assessments

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Overview of the Internship Activities

The Blackbucks Long Term Internship (LT 2026) is designed to prepare students for industry readiness through a structured approach involving **technical foundation strengthening**, **company-aligned mock assessments**, and **employability evaluations**.
During December 2025, the program focused on completing pending pre-requisite topics and periodically evaluating students' progress through mock tests and employability assessments.

Module 1: Pre-Requisites Learning (Skill Foundation Module)

Duration

- Conducted across **Weeks 01 to 04**
- Multiple days each week based on pending course requirements

Description

This module focused on strengthening students' **core technical and conceptual foundations** required for successful participation in the internship and future industry roles. Students continued learning pending syllabus topics aligned with their selected domain.

Key Learning Outcomes

- Completion of balance technical topics from the assigned course
- Improved conceptual clarity and problem-solving skills

- Readiness for company-level assessments and real-world tasks
- Reinforcement of fundamentals required for advanced internship activities

Mode

- Self-paced and guided learning
- Progress monitored through scheduled assessments

Module 2: Company-Based Mock Tests

Schedule

- **Mock Test 01:** 9th December 2025
- **Mock Test 02:** 16th December 2025
- **Mock Test 03:** 23rd December 2025
- **Mock Test 04:** 30th December 2025
- **Timing:** 6:00 PM – 11:00 PM

Description

Company-based mock tests were conducted to simulate **real corporate hiring assessments**. These tests were designed based on actual company recruitment patterns to familiarize students with industry expectations.

Key Objectives

- Evaluate technical, aptitude, and logical reasoning skills
- Prepare students for real placement and internship tests
- Improve time management and assessment handling skills
- Identify skill gaps and areas for improvement

Outcome

Students gained hands-on experience with company-style evaluations, enhancing their confidence and assessment readiness.

Module 3: Employability Assessments

Schedule

- **Assessment 01:** 13th December 2025
- **Assessment 02:** 20th December 2025

- **Assessment 03:** 27th December 2025
- **Assessment 04:** 3rd January 2026
- **Timing:** 6:00 PM – 11:00 PM

Description

Employability assessments were conducted to measure students' overall **job-readiness**, beyond technical skills. These assessments focused on aptitude, analytical thinking, communication ability, and professional awareness.

Key Evaluation Areas

- Problem-solving and analytical ability
- Logical reasoning and aptitude skills
- Industry readiness and employability parameters
- Performance consistency across assessments

Outcome

The assessments provided a structured benchmark of student readiness and helped track progressive improvement throughout the internship period.

Overall Internship Outcomes (December 2025)

- Successful continuation and completion of pending pre-requisite learning modules
- Exposure to **four company-based mock tests**
- Participation in **four employability assessments**
- Enhanced technical competency and assessment confidence
- Improved industry alignment and placement preparedness

Conclusion

The December 2025 phase of the Blackbucks Long Term Internship (LT 2026) provided a structured and outcome-oriented learning experience. Through continuous learning, regular mock testing, and employability assessments, students demonstrated progressive improvement in both technical and professional competencies.
This month served as a strong foundation for advanced internship activities and real-world industry engagement in subsequent phases.

Phase 02: Core Concept Phase (Week 05 – 12)

Duration: Week 03 onwards (February 2026)

Phase Overview

The Core Concept Phase focused on modern **data engineering technologies, big data frameworks, data pipeline development, and cloud-based data processing.** This phase strengthened practical data engineering skills and enhanced placement readiness through continuous assessments and hands-on learning.

Week 05: Data Storage, Formats & ETL Fundamentals

Duration: 19th January 2026 – 24th January 2026

Key Activities

- Placement Mock Test (3 Hours Duration)
- Session on Data Storage & Formats
- Assessment 02
- Placement Mock Test (3 Hours Duration)
- Session on ETL Basics
- Assessment 03

Learning Focus

- Structured and unstructured data concepts
- Data storage formats such as CSV, JSON, Parquet, and Avro
- Data schema and organization concepts
- Introduction to ETL (Extract, Transform, Load) processes

Week 05 Outcomes

- Understanding of different data storage formats
- Ability to understand basic ETL workflows
- Improved knowledge of data engineering foundations
- Enhanced readiness for placement-oriented assessments

Week 06: ETL Workflows & Big Data Foundations

Duration: 2nd February 2026 – 7th February 2026

Key Activities

- Placement Mock Test (3 Hours Duration)
- Assignment 2
- Session on ETL Basics
- Assessment
- Employability Test
- Session on Big Data Foundations

Learning Focus

- ETL pipeline design and data transformation techniques
- Introduction to big data ecosystems
- Understanding large-scale data processing concepts
- Data workflow and pipeline architecture

Week 06 Outcomes

- Ability to understand ETL pipeline architecture
- Basic knowledge of big data technologies
- Improved understanding of data processing workflows
- Strengthened analytical and technical readiness

Week 07: Hadoop Ecosystem & Distributed Data Processing

Duration: 9th February 2026 – 14th February 2026

Key Activities

- Session on Hadoop Fundamentals
- Assessment & Placement Mock Test
- Session on Hive & HBase Basics
- Assessment & Assignment 3
- Employability Test & Assessment
- Session on Apache Spark Overview

Learning Focus

- Hadoop architecture and distributed storage
- Data warehousing with Hive
- NoSQL database concepts with HBase
- Introduction to Apache Spark for distributed computing

Week 07 Outcomes

- Understanding of Hadoop ecosystem components
- Ability to work with distributed data storage systems
- Knowledge of Spark for large-scale data processing
- Improved big data analytical skills

Week 08: Spark Processing & Real-Time Data Streaming

Duration: 16th February 2026 – 21st February 2026

Key Activities

- Session on Spark SQL
- Assessment & Placement Mock Test
- Session on Spark Streaming
- Assessment & Assignment 4
- Employability Test
- Session on Apache Kafka Fundamentals

Learning Focus

- Querying large datasets using Spark SQL
- Real-time data processing using Spark Streaming
- Event-driven architecture with Apache Kafka
- Data streaming pipeline concepts

Week 08 Outcomes

- Ability to process large datasets using Spark
- Understanding of real-time data streaming systems
- Knowledge of Kafka for distributed messaging systems
- Improved real-time data processing capabilities

Week 09: Workflow Orchestration & Data Lake Architecture

Duration: 23rd February 2026 – 28th February 2026

Key Activities

- Session on Airflow Basics
- Assessment
- Placement Mock Test
- Session on Data Lake Architecture
- Session on Apache Kafka Fundamentals
- Employability Test & Assessment

Learning Focus

- Workflow scheduling and orchestration using Apache Airflow
- Data lake concepts and architecture
- Managing large-scale data pipelines
- Integration of messaging systems with data pipelines

Week 09 Outcomes

- Ability to schedule and manage ETL workflows
- Understanding of data lake storage architecture
- Improved pipeline orchestration knowledge
- Strengthened practical data engineering skills

Week 10: Cloud Data Engineering & Pipeline Monitoring

Duration: 2nd March 2026 – 7th March 2026

Key Activities

- Session on Cloud Data Engineering
- Assessment
- Placement Mock Test
- Session on ETL with PySpark
- Employability Test & Assessment
- Session on Monitoring & Logging

Learning Focus

- Cloud-based data engineering architectures
- Distributed data processing using PySpark
- Monitoring data pipelines and logging processes
- Ensuring reliability and performance of data workflows

Week 10 Outcomes

- Ability to build ETL pipelines using PySpark
- Understanding cloud data engineering environments
- Knowledge of monitoring and logging tools for pipelines
- Strengthened industry-ready data engineering skills

Week 11: Workflow Orchestration & Data Pipeline Architecture

Duration: 23rd February 2026 – 28th February 2026

Key Activities

- Session on Airflow Basics
- Assessment
- Placement Mock Test
- Session on Data Lake Architecture
- Session on Apache Kafka Fundamentals
- Employability Test & Assessment

Learning Focus

- Workflow orchestration using Apache Airflow
- Designing and managing data pipelines
- Data lake architecture and storage strategies
- Real-time data streaming using Apache Kafka
- Continuous assessment and placement preparation

Week 11 Outcomes

- Ability to design and manage data pipelines using Airflow
- Understanding of scalable data lake architectures
- Knowledge of real-time data streaming systems
- Improved data engineering and analytical skills

Week 12: Cloud Data Engineering & Pipeline Optimization

Duration: 2nd March 2026 – 7th March 2026

Key Activities

- Session on Cloud Data Engineering
- Assessment
- Placement Mock Test
- Session on ETL with PySpark
- Employability Test & Assessment
- Session on Monitoring & Logging

Learning Focus

- Cloud-based data engineering concepts and architectures
- Building ETL pipelines using PySpark
- Distributed data processing in cloud environments
- Monitoring and logging of data pipelines
- Ensuring reliability and performance of data workflows

Week 12 Outcomes

- Ability to implement ETL pipelines using PySpark
- Understanding of cloud data engineering workflows
- Knowledge of monitoring and logging mechanisms
- Enhanced data engineering and industry readiness

Phase 03: Recap & Review Phase (Week 13 – 14)

Duration: Final Weeks (March 2026 onwards)

Phase Overview

The Recap & Review Phase focused on consolidating all Data Engineering and Big Data concepts learned throughout the program. Emphasis was on revising core technologies, strengthening weak areas, and preparing for real-world data pipeline and big data challenges along with placement readiness.

Week 13: Big Data Processing, Streaming & Cloud Integration

Duration: 9th March 2026 – 14th March 2026

Key Activities

- Session on Big Data Tools (Hadoop, Spark) & Data Pipelines
- Placement Mock Test (3 Hours Duration)
- Assessment & Assignment 6
- Session on Cloud Platforms (AWS / Azure / GCP) for Data Engineering
- Employability Test & Assessment
- Session on Data Streaming (Kafka / Real-time Processing)

Learning Focus

- Building scalable data pipelines using Hadoop and Spark
- Real-time data processing using streaming tools like Kafka
- Cloud-based data storage and processing (AWS, Azure, GCP)
- ETL (Extract, Transform, Load) processes and data integration
- Continuous assessment and placement preparation

Week 13 Outcomes

- Ability to design and implement scalable data pipelines
- Understanding of real-time data processing systems
- Knowledge of deploying data solutions on cloud platforms
- Strengthened data engineering and big data processing skills

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Week 14: Data Engineering Project Development & Implementation

Duration: 16th March 2026 – 21st March 2026

Key Activities

- Session on Project Phase
- Employability Test
- Placement Mock Test (3 Hours Duration)
- Assignment 5
- Session on Project Phase

Learning Focus

- Development of end-to-end data engineering projects
- Implementation of ETL pipelines and data workflows
- Applying big data tools and cloud integration in projects
- Data validation, debugging, and optimization
- Placement-oriented assessment practice

Week 14 Outcomes

- Ability to develop complete data engineering solutions
- Improved integration of data pipelines and cloud services
- Enhanced debugging and performance optimization skills
- Increased confidence and readiness for real-world roles

Phase 04: Project Implementation Phase (Week 15)

Duration: Mid March 2026 onwards

Phase Overview

The Project Implementation Phase focused on applying Data Engineering and Big Data skills through real-world projects, emphasizing pipeline development, scalability, and data-driven solutions.

Week 15: Data Engineering Project Finalization & Evaluation

Duration: 23rd March 2026 – 28th March 2026

Key Activities

- Session on Project Phase
- Employability Test
- Placement Mock Test (3 Hours Duration)
- Assignment 5
- Session on Project Phase

Learning Focus

- Finalization of data pipelines and workflows
- Performance tuning and optimization of big data systems
- Integration of multiple data sources and processing layers
- Data validation, testing, and refinement
- Preparation for technical interviews and assessments

Week 15 Outcomes

- Ability to finalize and deliver complete data engineering projects
- Improved performance optimization and debugging skills
- Strong understanding of end-to-end data pipeline development
- Enhanced confidence and readiness for industry roles

Phase 05: Final Phase (Week 16)

Duration: Week 16 (March 2026 – Ending Phase)

Phase Overview

The Final Phase focused on completing all documentation and submissions, ensuring a professional conclusion to the internship.

Key Focus Areas

- Final project report preparation and submission
- Documentation of all phases and learnings
- Portfolio and resume finalization
- Submission of internship deliverables
- Feedback and evaluation

Outcome

Students successfully completed and submitted their work, showcasing a well-structured portfolio aligned with Data Engineering and Big Data industry standards.

Overall Phase-Wise & Week-Wise Outcomes

- Structured progression from data storage fundamentals to advanced data pipeline development
- Continuous assessments and skill benchmarking
- Hands-on exposure to Hadoop, Spark, Kafka, and Airflow
- Practical experience with ETL workflows and cloud data engineering
- Strong placement-oriented preparation and improved professional readiness

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Lesson Plan Link :
https://taptap.blackbucks.me/lessonPlan/?lessonPlanId=73226&testType=collegeLessonPlan

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Conclusion

The phase-wise structure of the **Blackbucks Long Term Internship 2026 – Data Engineering & Big Data Specialist** ensured systematic skill development. From the **Fundamental Phase**

(Phase 02) to the **Core Concept Phase (Phase 03)**, students built strong analytical foundations and practical expertise.

The **Recap & Review Phase (Phase 04)** strengthened understanding through revision and interview preparation, while the **Project Implementation Phase (Phase 05)** provided hands-on experience with real-world projects, ensuring complete industry readiness.